eraging over n first (Assume n ≥ 0)

In[14]:=
```
W[n_, y_, σy_] =
 Sqrt[(ω^2 + σy^2) / (2 π)] Exp[- (ω^2 + σy^2) y^2 / 2] (a n + c (ω^2 + σy^2) ω^2 y)
```
raíz cuadrada · exponencial

Out[14]=

$$\frac{e^{\frac{1}{2} y^2 \left(-\sigma y^2-\omega^2\right)} \sqrt{\sigma y^2+\omega^2}\, \left(a\, n + c\, y\, \omega^2 \left(\sigma y^2+\omega^2\right)\right)}{\sqrt{2 \pi}}$$

In[65]:=
```
G[y_, σy_] = Assuming[σn > 0, FullSimplify[
   Integrate[W[n, y, σy] / Sqrt[π σn^2 / 2] Exp[-n^2 / (2 σn^2)], {n, 0, ∞}]]]
```
asumiendo · simplifica completamente · integra · raíz cuadrada · exponencial

Out[65]=

$$\frac{e^{-\frac{1}{2} y^2 \left(\sigma y^2+\omega^2\right)} \sqrt{\sigma y^2+\omega^2}\, \left(2\, a\, \sigma n + c \sqrt{2 \pi}\, y\, \omega^2 \left(\sigma y^2+\omega^2\right)\right)}{2 \pi}$$

In[66]:=
```
D[Log[G[y, σy]], y] // FullSimplify
```
logaritmo · simplifica completamente

Out[66]=

$$-\frac{\left(\sigma y^2+\omega^2\right) \left(2\, a\, y\, \sigma n + c \sqrt{2 \pi}\, \omega^2 \left(-1 + y^2 \left(\sigma y^2+\omega^2\right)\right)\right)}{2\, a\, \sigma n + c \sqrt{2 \pi}\, y\, \omega^2 \left(\sigma y^2+\omega^2\right)}$$

In[67]:=
```
D[Log[G[y, σy]], σy] // FullSimplify
```
logaritmo · simplifica completamente

Out[67]=

$$\sigma y \left(-y^2 + \frac{1}{\sigma y^2+\omega^2} + \frac{2\, c \sqrt{2 \pi}\, y\, \omega^2}{2\, a\, \sigma n + c \sqrt{2 \pi}\, y\, \omega^2 \left(\sigma y^2+\omega^2\right)}\right)$$

In[69]:=

$$\text{Solve}\left[\left\{-\frac{\left(\sigma y^2+\omega^2\right) \left(2\, a\, y\, \sigma n + c \sqrt{2 \pi}\, \omega^2 \left(-1 + y^2 \left(\sigma y^2+\omega^2\right)\right)\right)}{2\, a\, \sigma n + c \sqrt{2 \pi}\, y\, \omega^2 \left(\sigma y^2+\omega^2\right)} == 0,\right.\right.$$

$$\left.\left.\sigma y \left(-y^2 + \frac{1}{\sigma y^2+\omega^2} + \frac{2\, c \sqrt{2 \pi}\, y\, \omega^2}{2\, a\, \sigma n + c \sqrt{2 \pi}\, y\, \omega^2 \left(\sigma y^2+\omega^2\right)}\right) == 0\right\}, \{y, \sigma y\}\right] \text{ // FullSimplify}$$

resuelve · simplifica completamente

Out[69]=

$$\left\{\left\{y \to \frac{c \sqrt{2 \pi}\, \omega^2}{a\, \sigma n}, \sigma y \to -\frac{\sqrt{-\frac{a^2\, \sigma n^2}{2 \pi} - c^2\, \omega^6}}{c\, \omega^2}\right\}, \left\{y \to \frac{c \sqrt{2 \pi}\, \omega^2}{a\, \sigma n}, \sigma y \to \frac{\sqrt{-\frac{a^2\, \sigma n^2}{2 \pi} - c^2\, \omega^6}}{c\, \omega^2}\right\},\right.$$

$$\left.\left\{y \to -\frac{a\, \sigma n + \sqrt{a^2\, \sigma n^2 + 2\, c^2\, \pi\, \omega^6}}{c \sqrt{2 \pi}\, \omega^4}, \sigma y \to 0\right\}, \left\{y \to \frac{-a\, \sigma n + \sqrt{a^2\, \sigma n^2 + 2\, c^2\, \pi\, \omega^6}}{c \sqrt{2 \pi}\, \omega^4}, \sigma y \to 0\right\}\right\}$$

This solution is unfeasible (negative quantity in square roots or zero variance)

Averaging over n first (Assume n can be positive or negative)

In[70]:=
```
G1[y_, σy_] = Assuming[σn > 0, FullSimplify[
    Integrate[W[n, y, σy] / Sqrt[2 π σn^2] Exp[-n^2 / (2 σn^2)], {n, -∞, ∞}]]]
```
asumiendo | simplifica completamente | integra | raíz cuadrada | exponencial

Out[70]=

$$\frac{c\, e^{-\frac{1}{2} y^2 \left(\sigma y^2+\omega^2\right)}\, y\, \omega^2 \left(\sigma y^2+\omega^2\right)^{3/2}}{\sqrt{2\pi}}$$

In[71]:=
```
Solve[{D[Log[G1[y, σy]], y] == 0, D[Log[G1[y, σy]], σy] == 0}, {y, σy}] // FullSimplify
```
resuelve | ·· | logaritmo | ·· | logaritmo | simplifica completament

Out[71]=

$$\left\{\left\{y \to -\frac{1}{\omega},\ \sigma y \to 0\right\},\ \left\{y \to \frac{1}{\omega},\ \sigma y \to 0\right\}\right\}$$

This solution forces zero variance

Then we look for the equilibrium and average over n later (Assume n positive or negative)

In[21]:=
```
{D[Log[W[n, y, σy]], y], D[Log[W[n, y, σy]], σy]} // FullSimplify
```
·· | logaritmo | ·· | logaritmo | simplifica completamente

Out[21]=

$$\left\{-\frac{\left(\sigma y^2+\omega^2\right)\left(a\, n\, y + c\,\omega^2\left(-1+y^2\left(\sigma y^2+\omega^2\right)\right)\right)}{a\, n + c\, y\, \omega^2\left(\sigma y^2+\omega^2\right)},\ \sigma y\left(-y^2+\frac{1}{\sigma y^2+\omega^2}+\frac{2\, c\, y\, \omega^2}{a\, n + c\, y\, \omega^2\left(\sigma y^2+\omega^2\right)}\right)\right\}$$

In[22]:=
```
Assuming[σn > 0 && ω^2 > 0 && σy > 0,
  Integrate[(1 / (a n + c y ω^2 (σy^2 + ω^2))) / Sqrt[2 π σn^2] Exp[-n^2 / (2 σn^2)], {n, -∞, ∞}]]
```
asumiendo | integra | raíz cuadrada | exponencial

Out[22]=

$$\frac{e^{-\frac{c^2 y^2 \omega^4 \left(\sigma y^2+\omega^2\right)^2}{2 a^2 \sigma n^2}}\left(\pi\, \mathrm{Erfi}\left[\frac{c\, y\, \omega^2\left(\sigma y^2+\omega^2\right)}{\sqrt{2}\, a\, \sigma n}\right]-\mathrm{Log}\left[-\frac{a}{c\, y}\right]+\mathrm{Log}\left[\frac{a}{c\, y}\right]\right)}{a\sqrt{2\pi}\,\sigma n}\quad \text{if}\quad \frac{c\, y}{a}\notin \mathbb{R}$$

Numerical check

In[23]:=
```
NIntegrate[(-(σy^2 + ω^2) y + c (σy^2 + ω^2) ω^2 / (a n + c y ω^2 (σy^2 + ω^2))) / Sqrt[2 π σn^2] Exp[-n^2 / (2 σn^2)] /.
   {σn → 1, c → 0.5, a → 0.15, σy → 0.05, y → 1.75, ω → 1.3}, {n, -∞, ∞}]
```
integra numéricamente | raíz cuadrada | exponencial

Out[23]=

-2.38837

In[24]:=

$$\frac{e^{-\frac{c^2 y^2 \omega^4 (\sigma y^2+\omega^2)^2}{2 a^2 \sigma n^2}} \left(\pi \,\text{Erfi}\left[\frac{c\, y\, \omega^2 (\sigma y^2+\omega^2)}{\sqrt{2}\, a\, \sigma n}\right]\right)}{a \sqrt{2\pi}\, \sigma n}\, c\, (\sigma y^2+\omega^2)\, \omega^2 - (\sigma y^2+\omega^2)\, y \;/.$$

$\{\sigma n \to 1, c \to 0.5, a \to 0.15, \sigma y \to 0.05, y \to 1.75, \omega \to 1.3\}$

Out[24]=

-2.38837

In[48]:= NIntegrate[
integra numéricamente

$$\left(-y^2/2 + \frac{1}{2(\sigma y^2+\omega^2)} + \frac{c\, y^2\, \omega^2}{a\, n + c\, y\, \omega^2 (\sigma y^2+\omega^2)}\right) \Big/ \text{Sqrt}[2\,\pi\,\sigma n\,\text{^}2]\ \text{Exp}[-n\,\text{^}2 / (2\,\sigma n\,\text{^}2)] \;/.$$

raíz cuadrada exponencial

$\{\sigma n \to 1, c \to 0.5, a \to 0.15, \sigma y \to 0.05, y \to 1.75, \omega \to 1.3\}, \{n, -\infty, \infty\}]$

Out[48]=

-0.198101

In[49]:=

$$-y^2/2 + \frac{1}{2(\sigma y^2+\omega^2)} + \frac{e^{-\frac{c^2 y^2 \omega^4 (\sigma y^2+\omega^2)^2}{2 a^2 \sigma n^2}} \left(\pi \,\text{Erfi}\left[\frac{c\, y\, \omega^2 (\sigma y^2+\omega^2)}{\sqrt{2}\, a\, \sigma n}\right]\right)}{a \sqrt{2\pi}\, \sigma n}\, c\, y^2\, \omega^2 \;/.$$

$\{\sigma n \to 1, c \to 0.5, a \to 0.15, \sigma y \to 0.05, y \to 1.75, \omega \to 1.3\}$

Out[49]=

-0.198101

Solve the system of equations by eliminating the Erfi function from the first equation
resuelve función error imaginaria

In[27]:=

$$\text{sr} = \text{Solve}\left[\frac{r}{a\sqrt{2\pi}\,\sigma n}\, c\, (\sigma y^2+\omega^2)\, \omega^2 - (\sigma y^2+\omega^2)\, y == 0,\ r\right] \,//\, \text{Flatten}$$

resuelve aplana

Out[27]=

$$\left\{r \to \frac{a\sqrt{2\pi}\, y\, \sigma n}{c\, \omega^2}\right\}$$

In[50]:=

$$-y^2/2 + \frac{1}{2(\sigma y^2+\omega^2)} + \frac{r}{a\sqrt{2\pi}\,\sigma n}\, c\, y^2\, \omega^2 \;/.\; \text{sr} \,//\, \text{FullSimplify}$$

simplifica completamente

Out[50]=

$$-\frac{y^2}{2} + y^3 + \frac{1}{2(\sigma y^2+\omega^2)}$$

In[53]:=
$$\text{sy} = \text{Solve}\left[-\frac{y^2}{2} + y^3 + \frac{1}{2\left(\text{s2} + \omega^2\right)} == 0,\ \text{s2}\right]\ //\ \text{Flatten}$$

resuelve aplana

Out[53]=
$$\left\{\text{s2} \to \frac{-1 + y^2\,\omega^2 - 2\,y^3\,\omega^2}{y^2\,(-1 + 2\,y)}\right\}$$

Substitute the solution for bar {y} into one of the equations with Erfi

función error imaginaria

In[54]:=
$$\frac{e^{-\frac{c^2\,y^2\,\omega^4\,(\text{s2}+\omega^2)^2}{2\,a^2\,\sigma n^2}}\left(\pi\,\text{Erfi}\left[\frac{c\,y\,\omega^2\,(\text{s2}+\omega^2)}{\sqrt{2}\,a\,\sigma n}\right]\right)}{a\,\sqrt{2\,\pi}\,\sigma n}\,c\,\left(\text{s2}+\omega^2\right)\omega^2 - \left(\text{s2}+\omega^2\right)y\ /.\ \text{sy}\ //\ \text{FullSimplify}$$

simplifica completamente

Out[54]=
$$-\frac{a\,y\,\sigma n - \sqrt{2}\,c\,\omega^2\,\text{DawsonF}\left[\frac{c\,\omega^2}{\sqrt{2}\,\left(a\,y\,\sigma n - 2\,a\,y^2\,\sigma n\right)}\right]}{a\,y^2\,\sigma n - 2\,a\,y^3\,\sigma n}$$

Plot the nonlinear equation

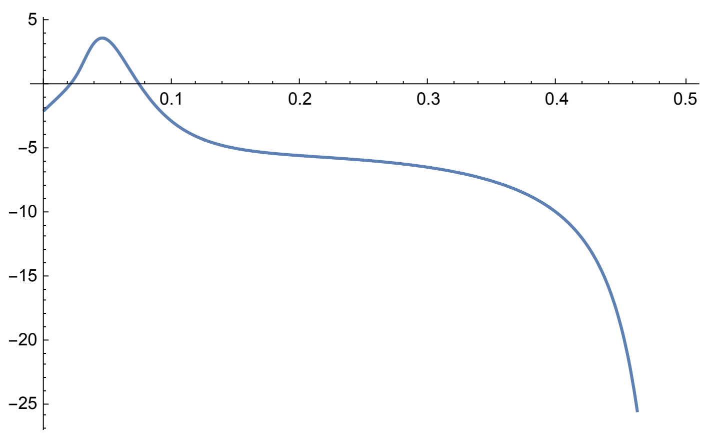
representación gráfica

$\xi = c\ /\ (a\,\sigma n)\ \omega^2$

In[84]:=
$$\text{Plot}\left[-\frac{y - \sqrt{2}\,\xi\,\text{DawsonF}\left[\frac{\xi}{\sqrt{2}\,\left(y - 2\,y^2\right)}\right]}{y^2 - 2\,y^3}\ /.\ \{\xi \to 0.1\},\ \{y, 0, 0.5\}\right]$$

representación gráfica

Out[84]=



In[82]:=
$$\text{syn} = \text{FindRoot}\left[-\frac{y - \sqrt{2}\,\xi\,\text{DawsonF}\left[\frac{\xi}{\sqrt{2}\,\left(y - 2\,y^2\right)}\right]}{y^2 - 2\,y^3} == 0\ /.\ \{\xi \to 0.1\},\ \{y, 0.001\}\right]$$

encuentra raíz

Out[82]=
$\{y \to 0.0198619\}$

Take c / (a σn) = 1

toma

In[83]:= $\frac{-1+y^2\,\omega^2-2\,y^3\,\omega^2}{y^2\,(-1+2\,y)}$ /. syn /. {ω → Sqrt[0.1]}
raíz cuadrada

Out[83]=
2639.65

Variance is positive, good
varianza

In[85]:= syn = FindRoot$\left[-\frac{y-\sqrt{2}\,\xi\,\text{DawsonF}\left[\frac{\xi}{\sqrt{2}\,(y-2\,y^2)}\right]}{y^2-2\,y^3}\right.$ == 0 /. {ξ → 0.1}, {y, 0.1}]
encuentra raíz

Out[85]=
{y → 0.073876}

In[86]:= $\frac{-1+y^2\,\omega^2-2\,y^3\,\omega^2}{y^2\,(-1+2\,y)}$ /. syn /. {ω → Sqrt[0.1]}
raíz cuadrada

Out[86]=
214.894

Variance is positive, good
varianza

Find the xi critical so that there are solutions for y. We solve
encuentra
the condition for the function taking the value zero at the maximum

In[95]:= sq = Solve$\left[-\frac{y-\sqrt{2}\,\xi\,q}{y^2-2\,y^3}\right.$ == 0, q] // Flatten
resuelve aplana

Out[95]=
$$\left\{q\to\frac{y}{\sqrt{2}\,\xi}\right\}$$

In[91]:= D$\left[-\frac{y-\sqrt{2}\,\xi\,\text{DawsonF}\left[\frac{\xi}{\sqrt{2}\,(y-2\,y^2)}\right]}{y^2-2\,y^3}\right.$, y] // FullSimplify
deriva simplifica completamente

Out[91]=
$$\frac{1}{(1-2\,y)^4\,y^5}\left(-y\left(1-6\,y+8\,y^2\right)\left(y^2\,(-1+2\,y)+\xi^2\right)+\sqrt{2}\,\xi\left(2\,(1-2\,y)^2\,y^2\,(-1+3\,y)+(1-4\,y)\,\xi^2\right)\text{DawsonF}\left[\frac{\xi}{\sqrt{2}\,\left(y-2\,y^2\right)}\right]\right)$$

In[96]:=

$$\frac{1}{(1-2\,y)^4\,y^5}\left(-y\left(1-6\,y+8\,y^2\right)\left(y^2\,(-1+2\,y)+\xi^2\right)+\sqrt{2}\,\xi\left(2\,(1-2\,y)^2\,y^2\,(-1+3\,y)+(1-4\,y)\,\xi^2\right)q\right)\ /.\ sq$$

Out[96]=

$$\frac{-y\left(1-6\,y+8\,y^2\right)\left(y^2\,(-1+2\,y)+\xi^2\right)+y\left(2\,(1-2\,y)^2\,y^2\,(-1+3\,y)+(1-4\,y)\,\xi^2\right)}{(1-2\,y)^4\,y^5}$$

In[98]:=

```
Solve[-y (1 - 6 y + 8 y^2) (y^2 (-1 + 2 y) + xi2) + y (2 (1 - 2 y)^2 y^2 (-1 + 3 y) + (1 - 4 y) xi2) == 0,
 xi2] // Flatten
```

resuelve aplana

Out[98]=

$$\left\{xi2\to\frac{y\left(-1+6\,y-12\,y^2+8\,y^3\right)}{2\,(-1+4\,y)}\right\}$$

In[101]:=

$$-y\left(1-6\,y+8\,y^2\right)\left(y^2\,(-1+2\,y)+\xi^2\right)+\sqrt{2}\,\xi\left(2\,(1-2\,y)^2\,y^2\,(-1+3\,y)+(1-4\,y)\,\xi^2\right)\mathrm{DawsonF}\left[\frac{\xi}{\sqrt{2}\left(y-2\,y^2\right)}\right]\ /.$$

$$\xi\to\mathrm{Sqrt}\left[\frac{y\left(-1+6\,y-12\,y^2+8\,y^3\right)}{2\,(-1+4\,y)}\right]\ //\ \mathrm{FullSimplify}$$

F de Dawson raíz cuadrada simplifica completamente

Out[101]=

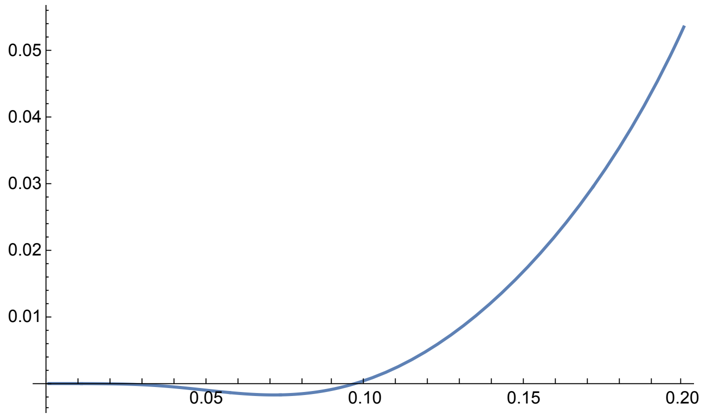
$$-\frac{1}{2}\,(1-2\,y)^2\,y\,(1+6\,y\,(-1+2\,y))\left(y-\sqrt{\frac{y\,(-1+2\,y)^3}{-1+4\,y}}\ \mathrm{DawsonF}\left[\frac{\sqrt{\frac{y\,(-1+2\,y)^3}{-1+4\,y}}}{2\,y-4\,y^2}\right]\right)$$

In[105]:=

$$\text{Plot}\left[y - \sqrt{\frac{y\,(-1+2\,y)^3}{-1+4\,y}}\ \text{DawsonF}\left[\frac{\sqrt{\frac{y\,(-1+2\,y)^3}{-1+4\,y}}}{2\,y-4\,y^2}\right],\ \{y,\ 0,\ 0.2\}\right]$$

representación gráfica   F de Dawson

Out[105]=



In[108]:=

$$\text{yc} = \text{FindRoot}\left[y - \sqrt{\frac{y\,(-1+2\,y)^3}{-1+4\,y}}\ \text{DawsonF}\left[\frac{\sqrt{\frac{y\,(-1+2\,y)^3}{-1+4\,y}}}{2\,y-4\,y^2}\right] == 0,\ \{y,\ 0.1\}\right]$$

encuentra raíz   F de Dawson

Out[108]=

$\{y \to 0.0964444\}$

At this value of y, the function touches the X axis and is negative otherwise (only one solution). Compute the critical xi

In[109]:=

$$\text{Sqrt}\left[\frac{y\,\left(-1+6\,y-12\,y^2+8\,y^3\right)}{2\,(-1+4\,y)}\right]\ /.\ \text{yc}$$

raíz cuadrada

Out[109]=

0.203171

In[110]:=

$$\text{Plot}\left[-\frac{y-\sqrt{2}\,\xi\,\text{DawsonF}\left[\frac{\xi}{\sqrt{2}\,\left(y-2\,y^2\right)}\right]}{y^2-2\,y^3}\;/.\;\{\xi\to 0.20317055115567076\},\,\{y,\,0,\,0.5\}\right]$$

representación gráfica

Out[110]=